



LINEAR ALGEBRA AND ITS APPLICATIONS

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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Supplementary problems

1. Find PA=LU and PA=LDU for $A = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

$$A = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow[R_3 - 1/3 R_1]{R_2 - 2R_1} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 4/3 & 2 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$PA = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 1 & 2 \\ 6 & -2 & 0 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - 1/3 R_1} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix} = U$$

$$PA = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4/3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{pmatrix}$$

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MATRICES AND GAUSSIAN ELIMINATION:



2. For what values of a and b does the following system have (i) a trivial solution

(ii) Infinitely many solutions.

$$\begin{aligned}x + 2y + 3z &= 0 \\ -x - 2y + az &= 0 \\ 2x + by + 6z &= 0\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & a \\ 2 & b & 6 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & a+3 \\ 0 & b-4 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & b-4 & 0 \\ 0 & 0 & a+3 \end{pmatrix}$$

This system will have only either a trivial solution or infinitely many solutions. It will have

(i) **a trivial solution** if $a \neq -3$ & $b \neq 4$ then $r(A)=3=n$

(ii) **Infinitely many solutions** if $a = -3$ or $b = 4$ or *both* then $r(A)$ will be 2 or 1 respectively.

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GAUSSIAN ELIMINATION:

3. Check for consistency and solve the following system of equations if consistent. Also discuss its rank:

$$\begin{array}{l} 3x + y + 2z = 3 \\ 2x - 3y - z = -3 \\ x + 2y + z = 4 \end{array} \quad \begin{pmatrix} 3 & 1 & 2:3 \\ 2 & -3 & -1:-3 \\ 1 & 2 & 1:4 \end{pmatrix} \xrightarrow{\begin{array}{l} R_2 - \left(\frac{2}{3}\right)R_1 \\ R_3 - \left(\frac{1}{3}\right)R_1 \end{array}} \begin{pmatrix} 3 & 1 & 2:3 \\ 0 & -11/3 & -7/3:-5 \\ 0 & 5/3 & -1/3:3 \end{pmatrix}$$

$$\xrightarrow{R_3 + \left(\frac{5}{11}\right)R_2} \begin{pmatrix} 3 & 1 & 2:3 \\ 0 & -11/3 & -7/3:-5 \\ 0 & 0 & -24/33:8/11 \end{pmatrix} \Rightarrow \begin{cases} 3x + y + 2z = 3 \\ (-11/3)y - 7/3z = -5 \\ (-24/33)z = 8/11 \end{cases}$$

$r(A)=r(A:b)=3=n$ hence system is **consistent** and has **a unique solution**.

i.e **$(x, y, z)=(1, 2, -1)$** . Its rank is **3**.

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4. Find an LU and LDU factorization for A. What is the rank of A?

$$\begin{aligned}
 A &= \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{pmatrix} \xrightarrow[\substack{R_3 - R_1 \\ R_4 + 3R_1}]{R_2 + 2R_1} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{pmatrix} \xrightarrow[\substack{R_4 - 4R_2}]{R_3 + 3R_2} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{pmatrix} \\
 &\xrightarrow{R_4 - 2R_3} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \quad A = LU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \\
 A = LDU &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1/2 & 5/2 & -1 \\ 0 & 1 & 1/3 & 2/3 & -1 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

Rank of A is 4.

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5. Solve $Ax=b$ for x if $A=\begin{pmatrix} 1 & 0 & -2 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{pmatrix}$ and $b=\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$

$$[A:I]=\begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 2 & 1 & 3 & : & 0 & 1 & 0 \\ 4 & 2 & 5 & : & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_3-4R_1]{R_2-2R_1} \begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 0 & 1 & 7 & : & -2 & 1 & 0 \\ 0 & 2 & 13 & : & -4 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_3-2R_2} \begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 0 & 1 & 7 & : & -2 & 1 & 0 \\ 0 & 0 & -1 & : & 0 & -2 & 1 \end{pmatrix} \xrightarrow[R_2+7R_3]{R_1-2R_3} \begin{pmatrix} 1 & 0 & 0 & : & 1 & 4 & -2 \\ 0 & 1 & 0 & : & -2 & -13 & 7 \\ 0 & 0 & -1 & : & 0 & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{-R_3} \begin{pmatrix} 1 & 0 & 0 & : & 1 & 4 & -2 \\ 0 & 1 & 0 & : & -2 & -13 & 7 \\ 0 & 0 & 1 & : & 0 & 2 & -1 \end{pmatrix} \quad A^{-1}=\begin{pmatrix} 1 & 4 & -2 \\ -2 & -13 & 7 \\ 0 & 2 & -1 \end{pmatrix} \quad x=\begin{pmatrix} 0 \\ 4 \\ -1 \end{pmatrix}$$



THANK YOU

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